

Gregor Ethan Rubin Posadas, M.S., E.I., ENV SP

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EDUCATION

Doctor of Philosophy in Civil and Environmental Engineering In Progress (Expected 2031)
University of California, Berkeley Berkeley, California
Environmental Engineering Program | Designated Emphasis in Development Engineering
Advisor: Dr. Kara L. Nelson | Co-advisor: Dr. Maya Carrasquillo

Master of Science in Civil Engineering December 2025
Boise State University Boise, Idaho
Thesis: Evaluation of High-Concentration Dissolved Oxygen Delivery on Wastewater Treatment Plant Performance
GPA: 4.0 | Tau Beta Pi Engineering Honor Society; Chi Epsilon Civil Engineering Honor Society

Bachelor of Science in Civil Engineering December 2023
Boise State University Boise, Idaho
Cum Laude | GPA: 3.74 | Dean's List (4 semesters)

RESEARCH EXPERIENCE

Research Fellow, Water Infrastructure and Policy September 2025 – August 2026
Water Finance Exchange (WFX), Inc. Washington, D.C. / Remote

- Conducted research on the adoption and implementation of new technologies for water infrastructure systems in small and rural communities across the United States.
- Performed purposive semi-structured stakeholder interviews and matrix coding analysis to surface barriers to effective water infrastructure-related policy implementation.
- Analyzed Aquatic Resource Alteration Permits (ARAPs) for water systems in Tennessee's Duck River watershed to identify opportunities for utility regionalization and consolidation.
- Developed a regionalization cost-estimate analysis by reviewing historical State Revolving Fund (SRF) funding requests from water systems across Tennessee, benchmarking the cost of infrastructure such as water treatment plant expansions, pumps, water lines, and water meters.

Research Manager February 2026 – August 2026
Boise State University, Department of Civil Engineering Remote

- Supervised research projects for Dr. Sondra Miller's research group, coordinating priorities and timelines across multiple concurrent studies.
- Mentored graduate and undergraduate students in research writing and the development of conference presentations.
- Developed grant proposals, data analysis reports, and peer-reviewed research publications.

Graduate Research Assistant December 2023 – December 2025
Boise State University, Department of Civil Engineering Boise, Idaho

- Performed comprehensive wastewater sampling and analysis (BOD, COD, nitrogen compounds, phosphorus, TSS, ATP) on a fully operational treatment facility.
- Conducted feasibility testing on novel retrofittable wastewater technology designed to improve bioreactor oxygenation and contaminant removal.
- Developed statistical and machine-learning approaches to evaluate relationships among dissolved oxygen, biomass activity, and sludge settleability.
- Mentored project-track graduate students through the journal publication process.

Undergraduate Research Assistant

August 2023 – December 2023

Boise State University, Department of Civil Engineering

Boise, Idaho

- Developed Python programs to support environmental laboratory experiments, including data analysis for settling column tests.
- Managed day-to-day operations of the Boise State University Environmental Engineering lab, maintaining equipment, safety protocols, and inventory.

TEACHING EXPERIENCE

Lab Instructor — CE 323: Environmental and Fluids Laboratory

August 2025 – December 2025

Boise State University, Department of Civil Engineering

Boise, Idaho

- Led hands-on lab sessions on water and wastewater quality testing (BOD, turbidity, pH, dissolved oxygen) and fluid mechanics.
- Evaluated lab reports and student performance to ensure mastery of experimental methods and engineering analysis.
- Taught students to analyze and interpret experimental data using statistical and graphical methods.

Teaching Assistant — CE 220: Coupled Environmental Processes

January 2025 – April 2025

Boise State University, Department of Civil Engineering

Boise, Idaho

- Graded assignments and exams, providing constructive feedback to improve student comprehension and analytical skills.
- Hosted weekly office hours offering one-on-one and small-group support on water and wastewater treatment concepts.

Guest Lecturer — CE 320: Environmental Engineering

October 2024

Boise State University, Department of Civil Engineering

Boise, Idaho

- Introduced methods for evaluating particle settling velocity distributions and constructing isoconcentration contour plots using beginner-friendly Python via Google Colab.

PUBLICATIONS

Peer-reviewed journal articles and conference proceedings (Posadas, G. bolded):

Posadas, G., & Bonacquist-Currin, M. (2026). *Barriers to technology adoption in small and rural water systems: A multi-stakeholder interview study.* Manuscript in preparation for submission to Engineering Research Express.

Henson, R. L., **Posadas, G., & Miller, S. M. (2026).** *Toward ATP-informed aeration control in wastewater treatment: A perspective.* Environmental Research Communications.

<https://doi.org/10.1088/2515-7620/aeb0f8>

Posadas, G., & Miller, S. (2026). *Wastewater treatment in Idaho in the face of emerging contaminants: Policy lag, utility burdens, and infrastructure readiness.* Environmental Research Communications.

<https://doi.org/10.1088/2515-7620/ae420a>

Posadas, G., Davis, K., & Miller, S. (2025). *High-concentration dissolved oxygen delivery in wastewater treatment: Barriers between innovation and adoption.* Environmental Research: Water.

<https://doi.org/10.1088/3033-4942/adffab>

Posadas, G., Davis, K., & Miller, S. (2025). *Liquid and solid phase effects of elevated bioreactor dissolved oxygen.* Proceedings of the 13th World Congress on Water Resources and Environment (EWRA 2025).

Posadas, G., Davis, K., & Miller, S. (2025). *Assessing the impacts of a novel dissolved oxygen infusion system on wastewater treatment plant performance.* Environmental Research Communications.

<https://doi.org/10.1088/2515-7620/addbaf>

Posadas, G., Baker, J. C., Ryan, A. S., & Miller, S. M. (2025). *Resiliency through applied research into emerging constituents*. Water Resources Management. <https://doi.org/10.1007/s11269-025-04140-7>

Posadas, G., & Miller, S. (2025). *Dissolved oxygen influences on sludge settleability: Mediating role of the volatile-to-total suspended solids ratio*. Accepted for publication in the 10th Asia Conference on Environment and Sustainable Development (ACESD 2025) conference proceedings.

CONFERENCE PRESENTATIONS

BahaWatch: Low-cost, open-source, locally-operated flood sensing for Philippine communities (and beyond) October 2026

Anglo, N., **Posadas, G.**, & Groeschel, T. — IEEE Global Humanitarian Technology Conference (GHTC 2026), Boulder, CO — Poster Presentation

ATP-based monitoring to indicate microbial performance in wastewater treatment plants July 2026
Smith, L., **Posadas, G.**, Henson, R., & Miller, S. — 2026 Idaho Conference on Undergraduate Research (ICUR), Boise State University, Boise, ID — Poster Presentation

Understanding wastes from Idaho beverage production April 2026
Parten, J., **Posadas, G.**, & Miller, S. — 2026 Undergraduate Research Showcase (URS), Boise State University, Boise, ID — Poster Presentation

Real-time opto-electrochemical PFAS and pharmaceutical sensing in water March 2026
Miller, S. M., Crockett, L. M., **Posadas, G. R.**, Omotowa, B., & Campbell, K. A. — DoW Applied Innovation Workshop, Washington Hilton, Washington, D.C. — Poster Presentation

Dissolved oxygen influences on sludge settleability: Mediating role of the volatile-to-total suspended solids ratio November 2025
10th Asia Conference on Environment and Sustainable Development (ACESD 2025), Fukuoka, Japan — Oral Presentation

Assessing the impacts of a novel dissolved oxygen infusion system on wastewater treatment plant performance September 2025
Pacific Northwest Clean Water Association Annual Conference (PNCWA 2025), Portland, OR — Oral Presentation

Liquid and solid phase effects of elevated bioreactor dissolved oxygen June 2025
13th World Congress on Water Resources and Environment (EWRA 2025), Palermo, Italy — Oral Presentation

Evaluation of high-concentration dissolved oxygen delivery on wastewater treatment plant performance November 2024
S-STEM Scholars & PI Meeting (AAAS & NSF), Chicago, IL — Poster Presentation

HONORS & AWARDS

Chancellor's Fellowship August 2026
University of California, Berkeley

IOP Publishing Outstanding Reviewer Award March 2026
Institute of Physics (IOP) Publishing

Semi-finalist, Fulbright U.S. Student Program (Singapore) 2026
U.S. Department of State, Bureau of Educational and Cultural Affairs

Inaugural MLK Jr. Living Legacy Committee Graduation Stole Recipient May 2025
Boise State University MLK Jr. Living Legacy Committee

Spring 2025 Graduating Student Leader Award (Graduate) May 2025
Boise State University Student Involvement and Leadership Center

Catalyzing Advocacy in Science and Engineering (CASE) Workshop AAAS Sponsorship Award American Association for the Advancement of Science (AAAS)	March 2025
United States Army Achievement Medal (AAM) U.S. Army Reserve, 321st Engineer Battalion	September 2024
Martin Luther King Jr. and Meredyth Burns Scholar Boise State University	April 2024
Commitment to Community Award Nominee Boise State University Division of Student Affairs	April 2024
ASCE New Face of Civil Engineering — Collegiate Honoree American Society of Civil Engineers (ASCE)	February 2024
NSF S-STEM Stellar Engineering Students Graduate Program Scholar (SEnS GPS) Boise State University Department of Civil Engineering	December 2023
Graduating Student Leader Award (Undergraduate) Boise State University Student Involvement and Leadership Center	December 2023
Fall 2023 Dean's List with High Honors Boise State University College of Engineering	December 2023
Idaho Opportunity Scholar Idaho State Board of Education	July 2023
391st Engineer Company Commander's Coin U.S. Army Reserve, 391st Engineer Company	April 2023
Spring 2023 Dean's List with High Honors Boise State University College of Engineering	May 2023
Spring 2021 Dean's List with Honors Boise State University College of Engineering	May 2021
Donald E. and Edithe L. Barclay Civil Engineering Scholar Boise State University Department of Civil Engineering	April 2021
Fall 2019 Dean's List with Highest Honors Boise State University College of Engineering	December 2019

CERTIFICATIONS & LICENSES

Engineer Intern (E.I.) — Credential E-22940 Idaho State Board of Professional Engineers and Land Surveyors	January 2024
Envision Sustainability Professional (ENV SP) — Credential 59936 Institute for Sustainable Infrastructure	January 2024
IOP Trusted Reviewer Status Institute of Physics (IOP) Publishing	October 2025
Fundamentals of Engineering (FE) Exam — Passed NCEES	December 2023

ADDITIONAL PROFESSIONAL EXPERIENCE

Wastewater Data Analyst Consultant BioSolutions, U.S.A.	August 2025 – July 2026 Greenleaf, Idaho
Conduct comparative measurements of activated-sludge dissolved oxygen concentrations to analyze oxygenation/aeration technology.	

- Monitor treatment-unit process technologies' power consumption for efficiency analyses.

Community Climate Action Committee Member

February 2025 – December 2025

City of Boise

Boise, Idaho

- Provided input on city policies integrating sustainability, resilience, and climate adaptation into urban planning and infrastructure.
- Contributed to the city's Community Climate Action Plan and advocated for youth involvement in climate initiatives.

Combat Engineer

August 2019 – July 2025

U.S. Army Reserve, 391st Engineer Company

Boise, Idaho

- Performed engineering calculations for obstacle emplacement and explosive demolitions, ensuring precision and safety in military operations.
- Operated and maintained military vehicles (M113A1, HMMWV, AVLM/AVLB, HEMTT, LMTV) and trained soldiers on radio (SINCGARS) and navigation (DAGR) systems.

Marketing and Graphic Design Delegate

November 2019 – January 2025

Boise State University MLK Jr. Living Legacy Committee

Boise, Idaho

- Organized the annual MLK Day march in Boise and other community engagement events raising awareness of civil rights and social justice.
- Established organizational brand standards and designed a cohesive suite of digital and print marketing materials.

Air Quality Monitoring Short-Term Consultant

March 2024 – April 2024

ADEC Environmental

Meridian, Idaho

- Assessed a pilot air-quality monitoring technology by comparing measurement capability against established EPA methods.
- Conducted daily ambient air-quality monitoring and prepared particulate-matter (PM) samples for laboratory analysis.

Civil Engineering Intern

May 2023 – September 2023

Stantec

Boise, Idaho

- Developed operations and maintenance (O&M) manuals for municipal wastewater treatment plants, ensuring compliance with operational standards.
- Finalized construction drawings using AutoCAD, Autodesk Civil 3D, and Bluebeam Revu, and prepared Statements of Qualifications (SOQs).
- Designed databases to track project pursuits and prepared site-condition exhibits and background-study reviews.

Lead Graphic Designer

August 2020 – May 2023

Dice & D-Pads

Quezon City, Philippines

- Assigned responsibilities and weekly agendas for the graphic design team, ensuring clear delegation and efficient workflows.
- Produced marketing materials, videos, and social media content using Adobe Creative Suite, Canva, and DaVinci Resolve.

Facilities, Operations, and Maintenance (FOM) Student Assistant

May 2021 – January 2022

Boise State University

Boise, Idaho

- Reviewed and updated university building plans for the department's online building database.
- Liaised between building managers and maintenance personnel to resolve facilities issues.

Help Desk Zone Support Analyst

October 2019 – December 2019

Boise State University, Office of Information Technology (OIT)

Boise, Idaho

- Provided in-person IT support to faculty and students, troubleshooting classroom technology and campus-wide services.
- Maintained lectern systems and managed equipment loans while documenting activity in the IT ticketing system.

ACADEMIC & DESIGN PROJECTS

Sustainable Energy Simulation and Analysis for Boise State University January 2024 – May 2024
Evaluated campus solar energy potential using PVSkyetch; produced economic projections and CO₂ offset analysis in coordination with the Boise State Office of Sustainability.

Project Proposal Development — Clearwater Engineering August 2023 – December 2023
Led a team responding to an RFP for a fish-barrier removal program; secured engineering partners and ranked first of four competing teams.

Municipal Wastewater Treatment Plant Design January 2023 – May 2023
Designed a theoretical municipal WWTP discharging to the Big Wood River — pump station, grit chambers, clarifiers, and UV disinfection — under IDAPA, HI, and GLUMRB standards; authored the Preliminary Engineering Report.

ASCE Timber Strong Competition January 2023 – April 2023
Helped design and construct a two-story timber structure for the ASCE Student Symposium (Boise State Civil Engineering Club).

Subdivision Construction Drafting January 2021 – May 2021
Created construction drawings for a Western Idaho subdivision, including roadway design, sewer/water line placement, and grade planning (Autodesk Civil 3D).

SOFTWARE DEVELOPMENT PROJECTS

Wastewater Treatment Plant Statistical Framework and Visualization Tool February 2025
github.com/gregor-posadas/wwtp-network-visualization

- Built a Python-based tool for wastewater data analysis using multiple-correlation methods and interactive visualizations.

Settling Column Isoremoval Curve Generator December 2024
isoremoval.streamlit.app

- Developed an open-access Python learning tool that generates isoremoval curves from raw settling column data; adopted in an Environmental Engineering course.

PUBLIC SPEAKING & OUTREACH

Featured Pinoy Scientist May 2025
pinoyscientists.com/scientist/gregor-ethan-posadas/

- Wrote and published daily social media posts on Philippine infrastructure and the role of Filipino scientists and engineers in national development.

Student Keynote Speaker — Fall 2023 Asian and Pacific Islander (API) Stole Ceremony December 2023
Hosted by the Boise State Student Equity Center (SEC) Boise, Idaho

- Delivered a keynote reflecting on cultural belonging and Filipino heritage.

Storyteller — The Story Collider: Veterans in STEM November 2023
storycollider.org/stories/2024/3/6/full-circle-stories-about-going-back-to-the-start

- Shared a personal narrative on the intersection of military service and a career in engineering.

SERVICE & VOLUNTEERING

Student Volunteer ASME EFX Event — Boise State University (Tau Beta Pi)	April 2024 – Present
Student Volunteer — Tree Planting Arbor Day Foundation	April 2024
Volunteer Rake Up Boise (Boise State Tau Beta Pi, Idaho Gamma Chapter)	November 2023

PROFESSIONAL AFFILIATIONS & LEADERSHIP

American Association for the Advancement of Science (AAAS) Student Member	April 2024 – Present
American Society of Civil Engineers (ASCE) Student Member	October 2022 – Present
Chi Epsilon Civil Engineering Honor Society Vice President (2024–25); Recording Marshal (2023–24)	May 2023 – May 2025
Water Environment Federation (WEF) Student Member	August 2024 – May 2025
Tau Beta Pi Engineering Honor Society Chapter Secretary	April 2023 – May 2024
ASBSU Sustainability Committee Student Member At-Large	February 2024 – May 2024
Associated Students of Boise State University (ASBSU) Review Board Chief	February 2024 – May 2024

TECHNICAL SKILLS

Engineering & Design Software: AutoCAD, Autodesk Civil 3D, Navisworks, Autodesk BIM 360, ProjectWise, Bluebeam Revu, SketchUp, PVSkyetch

Programming & Data: Python, MATLAB, statistical data analysis, machine learning, data visualization

Laboratory & Analytical: Wastewater sampling and analysis (BOD, COD, TSS, nitrogen, phosphorus, ATP); DO/pH/temperature multi-probe field methods

Productivity & Creative: Microsoft Office (Excel, Word, PowerPoint), Adobe Creative Suite, Canva, DaVinci Resolve

Professional: Research and technical writing, proposal writing, project management, grant development, stakeholder engagement

LANGUAGES

English (native/bilingual) • Tagalog (native/bilingual)

REFERENCES

Available upon request.